Helwan University **Optical Communications**
Faculty of Engineering **3^{rd} year (comm.)**
Electro. & Comm. Eng. Dept.

Sheet 5

Question 1

A fiber Bragg grating assisted coupler is designed to block an incoming optical signal present at the input port of the device. When the fiber core refractive index is 1.6 and the grating period is 0.42 μm, determine the wavelength of the blocked signal.

Question 2

If a diffraction grating produces its third order (m = 3) bright spot at an angle of 78.4° for light of wavelength 681 nm, find

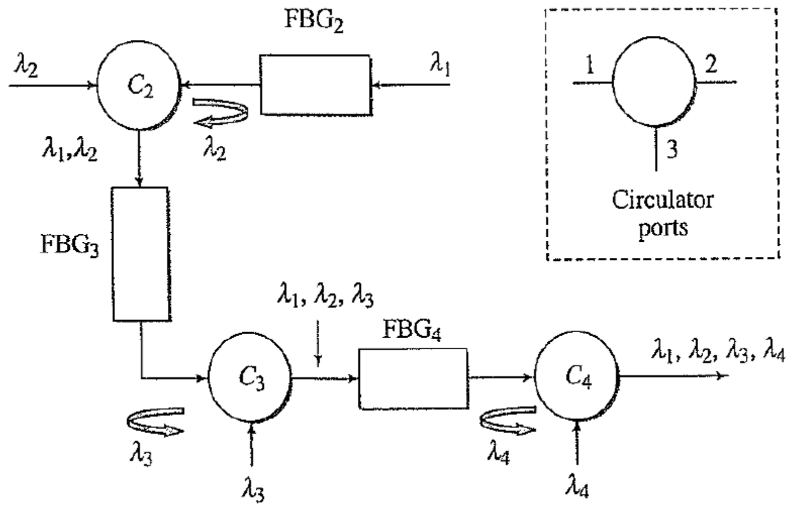
a) the number of slits per centimeter for the grating
b) the angular location of the first order and second order bright spots.
c) Will there be a fourth order bright spot? Explain.

Question 3

A diffraction grating with a 5.1×10^3 slit/cm is placed between a screen and light source. The distance between the grating and the screen is L=2 m. Please find the distance between first order maxima on the screen for the two wavelengths λ=514 nm and λ=635 nm.

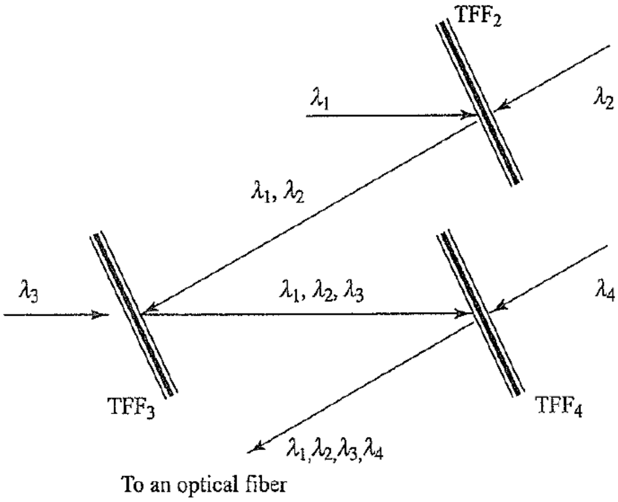
Question 4

Consider the wavelength multiplexer made from circulators and fiber Bragg gratings as shown in the following figure. Assume the input optical power of each wavelength $\lambda1$ through $\lambda4$ is 1mW. Let both the reflection loss and the throughput loss of each FBG be 0.25 dB, and let the insertion loss of a circulator be 0.6 dB. What is the power level of each wavelength $\lambda1$ through $\lambda4$ emerging from the final circulator C4?



Question 5

Consider the wavelength multiplexer made from a series of thin-film filters as shown in the following figure. Assume the input optical power of each wavelength $\lambda1$ through $\lambda4$ is 1mW. For each TFF, let the throughput loss equal 1 dB, and let the reflection loss be 0.4 dB. What is the power level of each wavelength λ1 through λ4 emerging from the final thin-film filter TFF4?



Question 6
Consider an array waveguide grating multiplexer that has the values for the operational variables listed in the following table.

a) Find the waveguide length difference.
b) Calculate the channel spacing Δν and the corresponding pass wavelength differential Δλ.

Table 10.12

Symbol	*Parameter*	*Value*
L_f	Focal length	9.38 mm
λ_0	Center wavelength	1554 nm
n_c	Array channel index	1.451
n_g	Group index for n_c	1.475
n_s	Slab waveguide index	1.453
x	Input/output waveguide spacing	25 μm
d	Grating waveguide spacing	25 μm
m	Diffraction order	118

Question 7
Design an AWG that can multiplex/demultiplex 16 WDM signals spaced 100 GHz apart in the 1.55 μm band. Your design must specify, among other things, the spacing between the input/output waveguides, the path length difference between successive arrayed waveguides, and the radius R of the grating circle.
Assume the refractive index of the input/output waveguides and the arrayed waveguides is 1.5. Note that the design may not be unique, and you may have to make reasonable choices for some of the parameters, which will in turn determine the rest of the parameters.